(a)

Explanation & Assumption

Task 1: Data Preparation & Product Revenue Analysis

In this assignment, I have the opportunity to simulate a Business Analyst for an online retailer based on the UK. The objective is to provide useful insights to the retailer via given two datasets by performing the use of R language.

The focus of Task 1 is to prepare data and conduct product revenue performance. The analysis begins with loading the datasets: `Project_orders`, which is the main table contains transaction sales data, and `project_product_info`, which contains product description.

Specifically, the Join operation is required to merge the Product Info table with `Stockcode` variable so that each transaction will include the corresponding product description. Therefore, the 1st Assumption is set to ensure that the Join operation is used to merge the two datasets, which is called `full` table. The next objective is to identify the Top 10 highest-selling products as `Top10_products` and Bottom 10 lowest-selling products as `Bottom10_products` based on the total revenue aggregated for each product. In addition, the Top 3 highest-selling products (`Top3_products_perc`) is also required to be analyzed in order to give better insights in depth for upcoming implications, which must be calculated as a percentage.

Finally, these results will be presented in tables and demonstrated by using plot functions to visualize them and later discuss the implications. In this case, I decided to use bar chart due to its clear, accurate, and easy-to-understand visual presentation. During the data inspection process, I realized that there should be more assumptions in this task to minimize any errors as much as possible. I observed that there is some invoice numbers begin with letter “C”, which means cancelled transactions, and the 2nd Assumption is needed to eliminate by removing Invoices records start with letter “C”. By doing this, I can remove records that do not represent complete sales records. The 3rd Assumption is also

introduced to meet the requirement of task 1, defining revenue variable by using the formula, which is `total_revenue`: Revenue = `Quantity x Unitprice`

This revenue variable is then updated as `total_revenue` to the `full`, supporting any upcoming calculations or analysis at the product level. While analyzing the lowest-selling products, some records are found to contain zero or negative revenue values. Therefore, I set the 4th Assumption of removing zero and negative revenue values within the "full" table, preventing errors from distorting the ranking of product performance.

Task 2: High Value Customer Analysis

In this task, the outputs are to Identify the top five customers (`Top5_customer`) based on their total spending across all transactions and conduct data analysis to understand their spending behavior (`Top5_behavior`), including which products they purchase the most. This insight will be later revealed through the two visualizations of (`Top5_customer`) and (`Top5_behavior`), aligning with the discussion in the interpretation section.

According to the task requirement, this analysis is based on `total_quantity` of products purchased by each customer. The results are later presented by using bar chart visualization through `geom_col` instead of `geom_bar` because `geom_bar` will automatically count the number of observations, which distorts the total revenue values of `y axis`. This allows a clear understanding of who contributes the most the retailer's revenue and their preferences for purchasing what kind of products.

During the initial inspection of the "full" table, I spotted that some `CustomerID` records contain missing values. Since these transactions cannot be linked to any specific customer, the 5th Assumption was drawn to remove all the missing values in the `CustomerID`. This ensures that the process of identifying top five value customers is based on valid `CustomerID`. When analyzing and plotting the customer purchasing behavior, the visualization resulted in overcrowded in `x axis` which is difficult to interpret. As a result, the 6th Assumption was set to select only top 5 products purchased by each of the top five

customers, highlighting the most significant purchasing patterns while keeping the visualization clear and readable.

Task 3: Global Market Analysis

The third objective of this assignment is to gain understanding about the product consumption in different countries, including both the domestic and international markets. By using data collected for sales in multiple countries, I am able to analyze the UK online retail shop's performance, derive insights on the global market and assess the market penetration of the retailer.

To start the analysis, the `total_revenue` generated in each Country is calculated by grouping the total revenue from transactions at the Country level, this value is later called `Country_Revenue` variable. This allows me to compare each country's share of total revenue and identified the product with the highest quantity sold in each market, labelled as `Highest_sellingproduct_country`.

In preparing the data for plotting total revenue by country, the 7th assumption specifies that only the top 22 countries were selected, excluding those with less than 7,000 pounds in revenue. This ensures that the `country_revenue_plot` remains clear and readable while retaining all countries with substantial and meaningful revenue contributions. Finally, the `Country_revenue_plot` is presented using bar plots to compare the total revenue between countries, supporting the retailer to grasp the store's performance in the regional markets and reveal potential opportunities in international markets.

Task 4: Temporal Analysis & Trends

Finally, the objective of this task is to reveal how sales change over time and to examine the overall performance trend. While inspecting temporal data to extract month and year transactions, the 8th Assumption of converting `InvoiceDate` variable into `POSIXct` format ensures that the `InvoiceDate` variable is interpreted correctly for further analysis and visualization. Once the time variables are extracted, `total_revenue` for each month is

calculated by aggregating revenue transactions based on the `year` and `month` data, which supports me to identify months with the highest sales clearly as `Highest_sales_month`.

However, in the process of analyzing product performance over time, I found that `Description` rows contain missing or empty product names. Since these records cannot be associated with a specific product, thus, the 9th Assumption is set to remove any missing or empty product descriptions from `Description` variable within the `full` table. After that, I identify products that are consistently among the lowest-selling products as `Lowest_product` by returning the smallest revenue in each year and month groups. In preparing temporal data for visualization, I created `year_month` variables to ensure that `geom_line` will plot the monthly sales performance correctly with x-axis in chronological order. As a result, I used line graphs for the time-series chart as visualization of `Monthly_sales_plot`, supporting the retailer to gain useful insights to notice any seasonal effects or patterns of temporal sales performance from Dec 2010 to Dec 2011.

Code

Task 1: Data Preparation & Product Revenue Analysis

I begin by loading and assigning the two datasets, `project_orders` and `project_product_info`, which contain the transaction records and product descriptions. These will be used throughout the analysis.

```
# load and read the datasets
# The two datasets are in Downloads, so "../" moves one level up from
Assignment to reach it.
orders  <- read.csv("../project_orders.csv")
product <- read.csv("../project_product_info.csv")
```

Next, I merge the two datasets using `StockCode` to create the `full` table.

```
# Assumption 1: Using Join function to merge the datasets
full <- left_join(orders, product, by = "StockCode")
```

Since invoice numbers beginning with “C” represent cancelled transactions, I remove these rows from the full table to avoid including invalid sales in the revenue calculations.

```
## I spotted that I should exclude Invoices starts with "C" in revenue
calculation
# => Assumption 2: Remove Invoices starts with "C" in revenue calculation
full <- full %>% filter(!str_starts(InvoiceNo, "C"))
```

I then create a new variable, `total_revenue`, calculated as $Quantity * UnitPrice$.

```
## Assumption 3: Create total revenue for each product
full <- full %>% mutate(total_revenue = Quantity * UnitPrice)
```

To analyse revenue at the product level, I aggregate revenue by `Description`.

```
# Create a data frame to make them easier to view
# Revenue is aggregated by Description to compute total revenue per product.
df <- data.frame(
  Description   = full$Description,
  total_revenue = full$total_revenue)

# Merge the duplicate invoice number
df_unique <- aggregate(total_revenue ~ Description, data = df, sum)

## To calculate top 10 highest and bottom 10 lowest-selling product ~ total revenue,
# I have to sort df in descending order
df_sorted <- df_unique %>% arrange(desc(total_revenue))
```

Some products have zero or negative revenue, so I remove these records in `df_unique` and update the `df_sorted` in descending order of total revenue.

```
## I noticed there are 0 revenue and negative revenue while testing Bottom 10 products
# => Assumption 4: remove 0 and negative revenue
any(df_unique$total_revenue < 0) # Check any 0 or negative revenue

## [1] TRUE

df_unique <- df_unique %>% filter(total_revenue>0)
any(df_unique$total_revenue < 0) # Re-check the Assumption. If it is false, then the filter is correct

## [1] FALSE

df_sorted <- df_unique %>% arrange(desc(total_revenue)) #re-run the code to update df_sorted
```

From the sorted results, I extract the `Top 10 highest-selling products`, `Bottom 10 lowest-selling products`, and `Top 3 highest-selling product in percentage`.

```
# Top 10 highest-selling product
Top10_products <- head(df_sorted,10)
kable(Top10_products)
```

Description	total_revenue
DOTCOM POSTAGE	206248.77
REGENCY CAKESTAND 3 TIER	174484.74

Description	total_revenue
PAPER CRAFT , LITTLE BIRDIE	168469.60
WHITE HANGING HEART T-LIGHT HOLDER	106471.28
PARTY BUNTING	99504.33
JUMBO BAG RED RETROSPOT	94340.05
mailout	85370.76
MEDIUM CERAMIC TOP STORAGE JAR	81700.92
Manual	78112.82
POSTAGE	78101.88

```
# Bottom 10 lowest-selling product
Bottom10_products <-tail(df_sorted, 10)

# Top 3 highest-selling product
Top3_products <- head(df_sorted,3)

# Top 3 highest-selling product in percentage
Top3_products_perc <- sum(Top3_products$total_revenue) /
sum(df_unique$total_revenue)*100
cat( "Top 3 highest-selling product in percentage is", Top3_products_perc,
"%.")

## Top 3 highest-selling product in percentage is 5.154116 %.
```

Presenting the above results in tables.

```
kable(Top10_products, caption = "Top 10 Highest-Selling Products")
```

Top 10 Highest-Selling Products

Description	total_revenue
DOTCOM POSTAGE	206248.77
REGENCY CAKESTAND 3 TIER	174484.74
PAPER CRAFT , LITTLE BIRDIE	168469.60
WHITE HANGING HEART T-LIGHT HOLDER	106471.28
PARTY BUNTING	99504.33
JUMBO BAG RED RETROSPOT	94340.05
mailout	85370.76
MEDIUM CERAMIC TOP STORAGE JAR	81700.92
Manual	78112.82
POSTAGE	78101.88

```
kable(Bottom10_products, caption = "Bottom 10 Lowest-Selling Products")
```

Bottom 10 Lowest-Selling Products

	Description	total_revenue
3717	BLACKCHRISTMAS TREE 30CM	1.250
3718	DUSTY PINK CHRISTMAS TREE 30CM	1.250
3719	ETCHED GLASS STAR TREE DECORATION	1.250
3720	ORANGE FELT VASE + FLOWERS	1.250
3721	PACK 4 FLOWER/BUTTERFLY PATCHES	1.250
3722	CAT WITH SUNGLASSES BLANK CARD	0.950
3723	HAPPY BIRTHDAY CARD TEDDY/CAKE	0.950
3724	PINK CRYSTAL GUITAR PHONE CHARM	0.850
3725	VINTAGE BLUE TINSEL REEL	0.840
3726	PADS TO MATCH ALL CUSHIONS	0.003

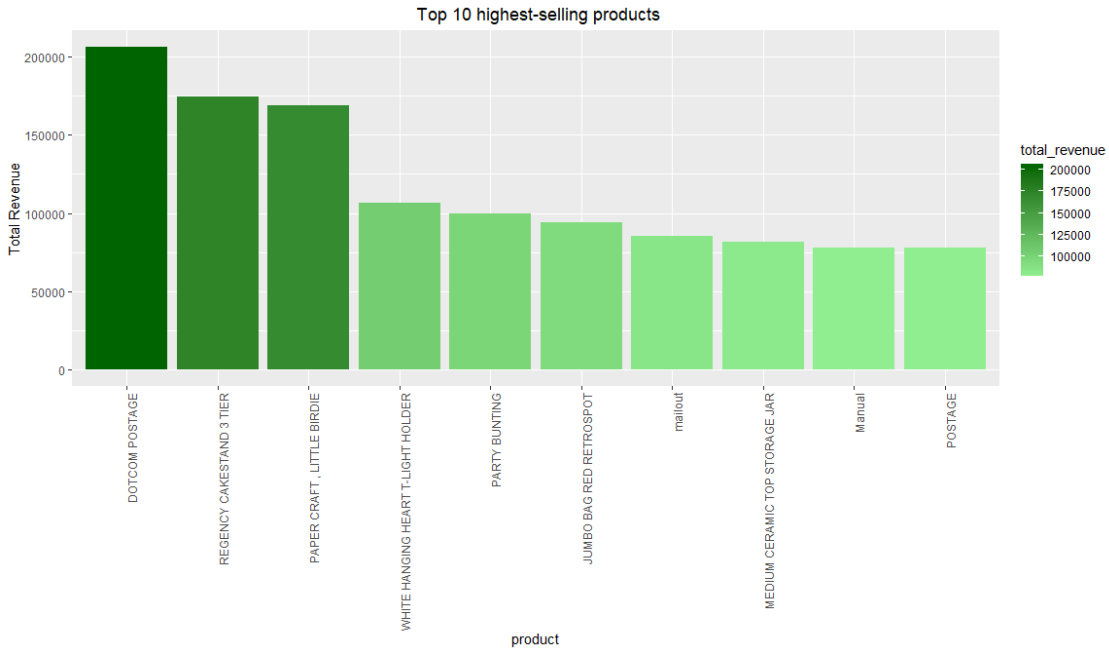
```
kable(Top3_products, caption = "Top 3 Products by Revenue")
```

Top 3 Products by Revenue

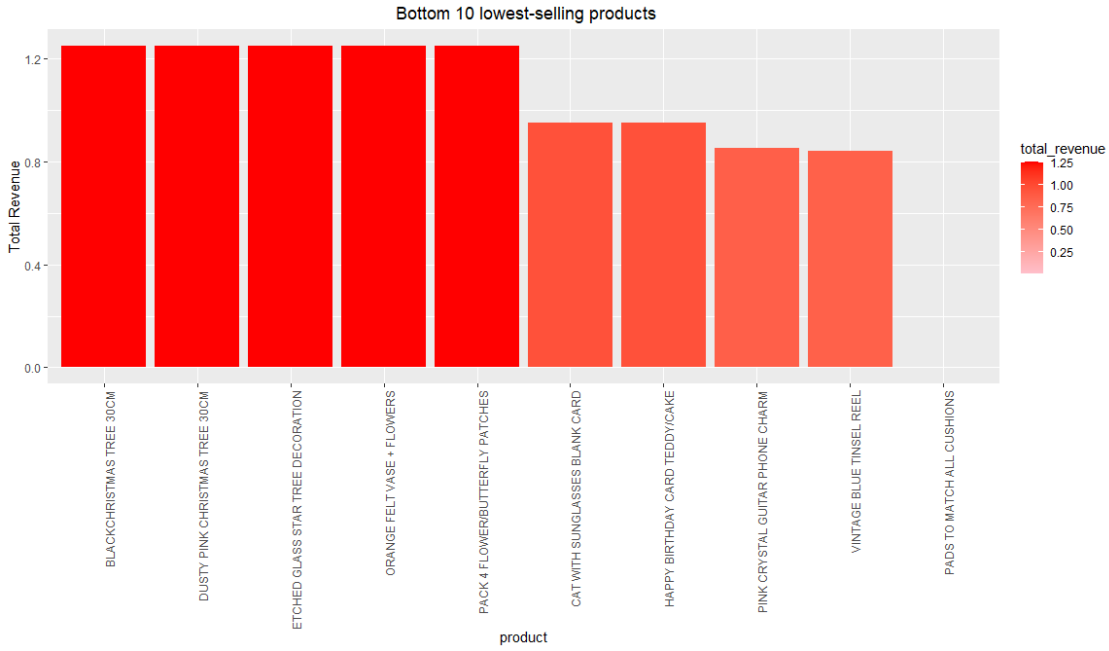
Description	total_revenue
DOTCOM POSTAGE	206248.8
REGENCY CAKESTAND 3 TIER	174484.7
PAPER CRAFT , LITTLE BIRDIE	168469.6

Finally, I visualise the revenue concentration using bar charts for both the `Top_10` and `Bottom_10` products.

```
# Using bar plots to visualize the revenue concentration
# Top 10 highest-selling products
Top10_bar <- ggplot(data= Top10_products, aes(x= reorder(Description, -
total_revenue), y= total_revenue, fill = total_revenue))+ geom_col() + labs(
title =" Top 10 highest-selling products", x= "product", y="Total Revenue") +
scale_fill_gradient(low="light green", high="dark green") +theme(axis.text.x=
element_text(angle=90, hjust=1)) +theme(plot.title= element_text(hjust =
0.5))
print(Top10_bar)
```




```r
# Bottom 10 lowest-selling products
Bottom10_bar <- ggplot(data= Bottom10_products, aes(x= reorder(Description, -total_revenue), y= total_revenue, fill = total_revenue))+ geom_col() + labs( title =" Bottom 10 lowest-selling products", x= "product", y="Total Revenue") + scale_fill_gradient(low="pink", high="red") +theme(axis.text.x= element_text(angle=90, hjust=1))+theme(plot.title= element_text(hjust = 0.5))
print(Bottom10_bar)
```



Task 2: High-Value Customer Analysis

I start by removing rows where CustomerID is missing, as these entries cannot be used for customer-level analysis.

```
any(is.na(full$CustomerID)) #Check missing value in Customer ID

## [1] TRUE

# Assumption 5: remove all the missing values in Customer ID
full <- full %>% filter(!is.na(CustomerID))
any(is.na(full$CustomerID)) # Re-check the missing values.

## [1] FALSE
```

Next, I calculate the total spending for each customer by summing `total_revenue` across all their transactions, and then identify the Top 5 highest-spending customers.

```
# Calculate customer spending
customer_spending <- full %>% group_by(CustomerID) %>%
summarise(total_spending=sum(total_revenue)) %>%
arrange(desc(total_spending))
#Identify the TOp 5 highest spending customers
Top5_customer <- head(customer_spending, 5)
kable(Top5_customer)
```

CustomerID	total_spending
14646	280206.0
18102	259657.3
17450	194550.8
16446	168472.5
14911	143825.1

To better understand the purchasing behaviour of these customers, I extract the detailed transaction records associated with the Top 5 customers.

```
#Create Top 5 customer spending with more details from full table for more
information
Top5_data <- full %>% filter(CustomerID %in% Top5_customer$CustomerID) %>%
select(InvoiceNo, StockCode, Quantity, InvoiceDate, UnitPrice, CustomerID,
Country, Description, total_revenue) %>% arrange(desc(Quantity))
```

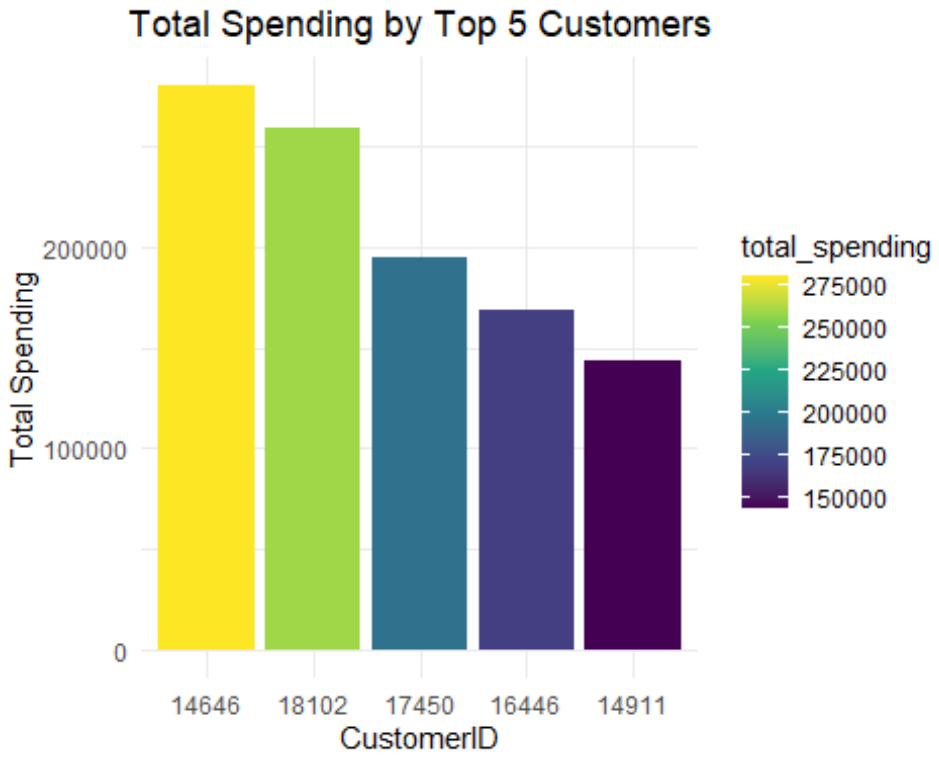
I summarise their purchasing behaviour by grouping `CustomerID` and `Description` variables.

```
## Create Top 5 Customer Behavior table based on Top5_data
# to analyse their purchasing behavior to identify
# which products they purchased the most (based on total quantity)
Top5_behavior <- Top5_data %>% group_by(CustomerID,Description) %>%
```

```
  summarise(total_quantity=sum(Quantity), total_revenue = sum(total_revenue),
.groups="drop") %>%
  arrange(CustomerID, desc(total_quantity), desc(total_revenue))
```

Then plotting the total spending of the Top 5 customers.

```
# Visualize the Top 5 customer and Top 5 behavior by bar chart
options(scipen=9999) # To make value of total_spending measurable
# Top 5 customer spending plot
Top5_customer_bar <- ggplot(Top5_customer, mapping=aes(x= reorder(CustomerID,
-total_spending), y= total_spending, fill= total_spending)) +geom_col() +
labs(title = "Total Spending by Top 5 Customers", x="CustomerID", y="Total
Spending") + scale_fill_viridis_c() +theme_minimal()+theme(plot.title=
element_text(hjust = 0.5))
print(Top5_customer_bar)
```

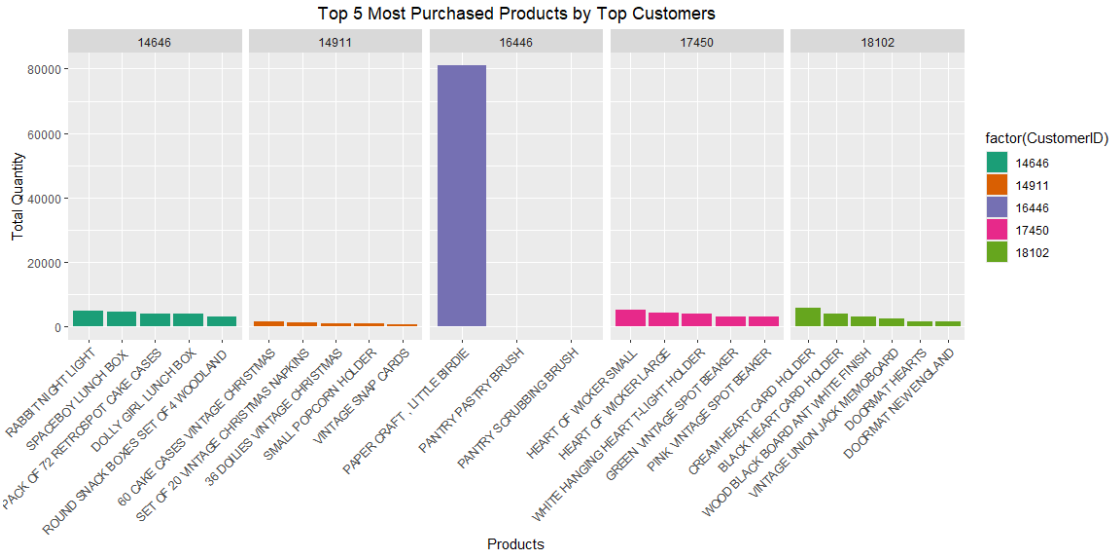


I select only Top 5 products for the upcoming of plotting Top 5 most purchased products for each customer.

```
## Assumption 6: Select only Top 5 products purchased by each Top customers
# Because the Top5_behaviour dataset has 2753 observations of products and
total_quantity,
# which results in overcrowded and unreable chart.
# Therefore, Assumption 6 helps clearly highlight customers' purchasing
behaviour.
Top5_products <- Top5_behavior %>% group_by(CustomerID) %>%
slice_max(order_by = total_quantity,n=5)
```

Finally, I plot the `Top 5 most purchased products` of each customer to illustrate their individual purchasing preferences.

```
# Top 5 customer behavior bar chart
Top5_behavior_plot <- ggplot(Top5_products, aes(x= reorder(Description, -
total_quantity), y=total_quantity, fill= factor(CustomerID))) +geom_col()
+facet_grid(~CustomerID, scale="free_x") + scale_fill_brewer(palette="Dark2")
+ labs(title = "Top 5 Most Purchased Products by Top Customers",
x="Products", y="Total Quantity")+theme(axis.text.x = element_text(angle =
45, hjust = 1),plot.title= element_text(hjust = 0.5))
print(Top5_behavior_plot)
```



Task 3: Global Market Analysis

I begin this task by calculating the `total_revenue` generated in each country.

```
# Revenue for each country
Country_Revenue <- full %>% group_by(Country) %>%
summarise(total_revenue=sum(total_revenue)) %>% arrange(desc(total_revenue))
kable(Country_Revenue)
```

Country	total_revenue
United Kingdom	7308391.55
Netherlands	285446.34
EIRE	265545.90
Germany	228867.14
France	209024.05
Australia	138521.31
Spain	61577.11
Switzerland	56443.95

Country	total_revenue
Belgium	41196.34
Sweden	38378.33
Japan	37416.37
Norway	36165.44
Portugal	33439.89
Finland	22546.08
Singapore	21279.29
Channel Islands	20450.44
Denmark	18955.34
Italy	17483.24
Cyprus	13590.38
Austria	10198.68
Poland	7334.65
Israel	7221.69
Greece	4760.52
Iceland	4310.00
Canada	3666.38
USA	3580.39
Malta	2725.59
Unspecified	2667.07
United Arab Emirates	1902.28
Lebanon	1693.88
Lithuania	1661.06
European Community	1300.25
Brazil	1143.60
RSA	1002.31
Czech Republic	826.74
Bahrain	548.40
Saudi Arabia	145.92

Then, I identify the single product with the highest `total_quantity` sold in each country, providing insight into regional product preferences.

```
# Identify single product with the highest quantity sold per country
Highest_sellingproduct_country <- full %>% group_by(Country, Description) %>%
summarise(total_quantity=sum(Quantity), .groups = "drop") %>%
slice_max(order_by = total_quantity, n=1, with_ties = FALSE) %>%
```

```
arrange(desc(total_quantity))
print(Highest_sellingproduct_country)

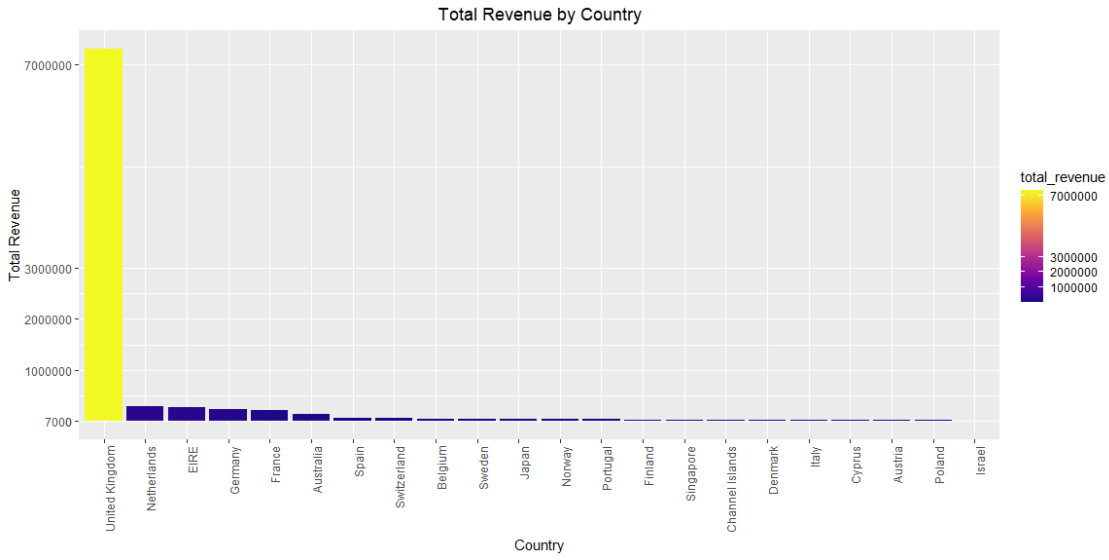
## # A tibble: 1 × 3
##   Country        Description                 total_quantity
##   <chr>          <chr>                                <int>
## 1 United Kingdom PAPER CRAFT , LITTLE BIRDIE          80995
```

Selecting only top 22 countries to improve readability and avoid overcrowded in x axis.

```
## Assumption 7: Select only top 22 countries by total revenue to avoid
overcrowded and improve readability.
# Moreover, the excluded countries have revenue less than 7000 pounds.
Country_Revenue_filtered <- Country_Revenue %>%
  slice_max(order_by = total_revenue, n = 22)
```

To visualize the differences in revenue across countries, I plot a bar chart showing total revenue by region.

```
# Visualization to compare revenue across regions
Country_revenue_plot <- ggplot(Country_Revenue_filtered , mapping=aes(x=
reorder(Country, -total_revenue), y= total_revenue, fill= total_revenue))
+geom_col() + labs(title = "Total Revenue by Country", x="Country", y="Total
Revenue") + scale_fill_viridis_c(option = "C", breaks = c(0, 7000, 1000000,
2000000, 3000000, 7000000)) +scale_y_continuous(breaks = c(0, 7000, 1000000,
2000000, 3000000, 7000000)) +theme(axis.text.x= element_text(angle=90,
hjust=1))+theme(plot.title= element_text(hjust = 0.5))
print(Country_revenue_plot)
```



Task 4: Temporal Analysis & Trends

I convert the `InvoiceDate` column into a proper datetime format so that extracting the year and month accurately for the time-series analysis would be accurate.

```
## Assumption 8: Convert InvoiceDate to POSIXct to ensure it is recognized as a datetime object
# so that year and month can be extracted correctly from "full" table
full$InvoiceDate <- as.POSIXct(full$InvoiceDate, format = "%d/%m/%Y %H:%M")
```

After converting the date, I extract the `year` and `month` data from the `InvoiceDate` variable to prepare the data for monthly revenue calculations.

```
# Extract months, years for each transaction from "full"
full <- full %>% mutate( year = as.numeric(format(InvoiceDate, "%Y")), month = as.numeric(format(InvoiceDate, "%m")))
```

I then calculate the total revenue for each month by grouping the data by `year` and `month` variables.

```
# Calculate total revenue for each month
monthly_revenue <- full %>% group_by(year, month) %>%
summarise(total_revenue=sum(total_revenue))
kable(monthly_revenue)
```

year	month	total_revenue
2010	12	572713.9
2011	1	569445.0
2011	2	447137.3
2011	3	595500.8
2011	4	469200.4
2011	5	678594.6
2011	6	661213.7
2011	7	600091.0
2011	8	645343.9
2011	9	952838.4
2011	10	1039318.8
2011	11	1161817.4
2011	12	518192.8

Next, I identify the months with the highest total revenue.

```
# Identify months with highest sales
Highest_sales_month <- monthly_revenue %>% slice_max(order_by = total_revenue, n=1)
print(Highest_sales_month)

## # A tibble: 2 × 3
## # Groups:   year [2]
##    year month total_revenue
```

```
##   <dbl> <dbl>         <dbl>
## 1  2010    12       572714.
## 2  2011    11      1161817.
```

After that, I analyse product performance over time by calculating monthly revenue for each product.

```
# Analyse product performance over time
monthly_product_revenue <- full %>% group_by(year, month, Description) %>%
summarise(product_revenue=sum(total_revenue))
any(full$Description == "" | is.na(full$Description)) # Check as if any
Description is empty product
```

```
## [1] TRUE
```

Some rows contain missing or empty product descriptions, so I remove these cases before recalculating monthly product revenue.

```
## Assumption 9: Remove Description rows is missing or have empty product
descriptions
full <- full %>% filter(!is.na(Description) & Description != "")
any(full$Description == "" | is.na(full$Description)) # re-Check as if any
Description is empty product
```

```
## [1] FALSE
```

```
monthly_product_revenue <- full %>% group_by(year, month, Description) %>%
  summarise(product_revenue=sum(total_revenue)) %>% arrange(year, month,
desc(product_revenue)) # Re-run the code
```

I then identify the products that are consistently lowest-selling in each month.

```
# Identify products consistently among the lowest-selling across multiple
months
Lowest_product <- monthly_product_revenue %>% group_by(year, month) %>%
slice_min(order_by=product_revenue, n=1)
kable(Lowest_product)
```

year	month	Description	product_revenue
2010	12	ASSORTED MONKEY SUCTION CUP HOOK	0.420
2010	12	METAL BASE FOR CANDLES	0.420
2010	12	SILICON CUBE 25W, BLUE	0.420
2010	12	SILICON STAR BULB BLUE	0.420
2011	1	ANGEL DECORATION STARS ON DRESS	0.420
2011	1	ASSORTED CIRCULAR MOBILE	0.420
2011	1	ASSTD RASTA KEY-CHAINS	0.420
2011	1	BLUE VICTORIAN FABRIC OVAL BOX	0.420

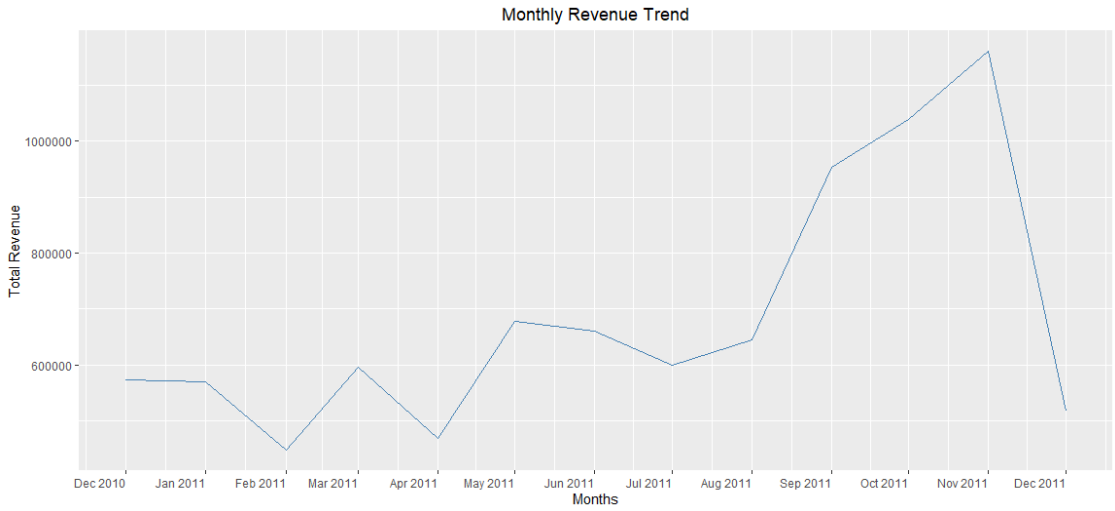
year	month	Description	product_revenue
2011	2	MAGIC DRAWING SLATE BAKE A CAKE	0.420
2011	2	MINI JIGSAW PURDEY	0.420
2011	3	BLACK FEATHER CHRISTMAS DECORATION	0.190
2011	4	PADS TO MATCH ALL CUSHIONS	0.001
2011	5	MINI JIGSAW GO TO THE FAIR	0.420
2011	5	MINI JIGSAW PURDEY	0.420
2011	6	ASSORTED COLOUR SUCTION CUP HOOK	0.190
2011	7	PADS TO MATCH ALL CUSHIONS	0.001
2011	8	EASTER DECORATION EGG BUNNY	0.390
2011	9	PADS TO MATCH ALL CUSHIONS	0.001
2011	10	HARDMAN MUG 3 ASSORTED	0.390
2011	11	HAPPY BIRTHDAY CARD STRIPEY TEDDY	0.190
2011	12	BLUE EGG SPOON	0.360

To prepare the data for a time-series plot, I create a year_month variable so that the x-axis is displayed in chronological order.

```
## Create a year_month variable so that R recognises the time variable as a Date object.
# This allows ggplot to correctly interpret the data as a time series and automatically
# plot the x-axis in chronological order.
monthly_revenue <- monthly_revenue %>% mutate( year_month = as.Date(paste(year,month,"01", sep="-"))) # add year_month for more detail
```

Finally, I visualize the monthly revenue trend using a line chart to show how sales fluctuate over time.

```
# Visualize the monthly sales trend using a time-series line chart
Monthly_sales_plot <- ggplot(monthly_revenue, mapping=aes(x= year_month, y= total_revenue)) +geom_line(color= 'steelblue') +scale_x_date(date_breaks= "1 month", date_labels = "%b %Y") + labs(title = "Monthly Revenue Trend", x="Months", y="Total Revenue") +theme(axis.text.x= element_text(hjust=1)) + theme(plot.title= element_text(hjust = 0.5))
print(Monthly_sales_plot)
```



Interpretation

Discuss the visualizations and their implications of each task.

Task 1: Data Preparation & Product Revenue Analysis

The `Top10_bar` shows the products that are generating the highest revenue. DOTCOM Postage witnesses largest revenue product with slightly above 200,000 pounds. The second and third top selling products are REGENTS CAKESTAND 3 TIER and PAPER CRAFT, LITTLE BIRDIE, which are sold for about 174,484 pounds and 168,469 pounds respectively. Moreover, the revenue in the Top 10 products shows a decline gradually, without an extreme drop, which indicates that although some of the products showed a stronger sales performance, there is no single product that overwhelmingly dominates the whole portfolio. Therefore, the revenue is not concentrated but instead fragmented. This can be further supported by the percentage of 5.15% contribution of Top 3 highest selling products.

In contrast, the `Bottom10_bar` represents products with extremely low revenues compared to the Top 10. The products in this group consisting of BLACKCHRISTMAS TREE 30CM, DUSTY PINK CHRISTMAS TEE 30CM, ETCHED GLASS STAR TREE DECORATION, ORANGE FELT VASE + FLOWERS, and PACK 4 FLOWER/BUTTERFLY PATCHES give revenue close to

1 pounds. The differences between these products in the `Bottom10_bar` are rather small, amounting to an estimate of 0.30 pounds, and their contribution to the sum of revenue is negligible.

From a business point of view, the diversified revenue structure lowers the dependency risk of the online retailer shop. Particularly, the UK online shop does not rely heavily on certain dominant products, implying that the supply disruption or a significant decline in demand for any product would have limited impact on the total revenue. Therefore, the greatest advantage to fragmented revenue is offering revenue stability.

However, there are trade-offs associated with this structure. For example, wider range of products may cause problems in promotional efforts for the shop owner, making management and product consideration an important element in marketing. Moreover, as shown from the `Bottom10_bar`, there are several products that generate extremely low revenue as compared to the other products, such as Vintage Blue Tinsel Reel with 0.840 pounds or Pads To Match All Cushions with 0.003 pounds. Meanwhile these products have minimal impact on the total revenue generated from them, the maintenance of a wide range of low revenue products may need to be done with enhanced efforts on inventory monitoring and product rationalization to ensure efficiency of operation.

Task 2: Discuss the importance of these customers and their preferences from a business perspective.

The `Top5_customer_bar` chart shows the total spending of the five most-spending customers within the given datasets. Customer 14646 has the highest spending with 280,206 pounds, followed closely by Customer 18102 with 259,657 pounds. The other three customers in the top 5 list are Customer 17450 with 194,551 pounds, Customer 16446 with 168,472 pounds and Customer 14911 with 143,825 pounds respectively. Even though the spending volume of these five customers may vary, they generate much higher revenue in comparison to the remaining customer base. This indicates that a relatively small group of customers drives a substantial portion of the retailer's overall revenue.

The `Top5_behavior_plot` gives further investigation of the purchasing behavior by indicating the five most frequently bought products by each customer. The visualization shows that although their contribution is substantial, their preferences regarding products are different, except for Customer 16466 who only purchases a large amount of PAPER CRAFT - LITTLE BRIDIE. Besides, Customer 14646 frequently purchases RABBIT NIGHT LIGHT, SPACEBOY LUNCH BOX, PACK OF 72 RETROSPOT CAKE GAME, DOLLY GIRL LUNCH BOX, and ROUND SNACK BOXES SET OF 4 WOODLAND, suggesting a reference of children's accessories, lunchware, and small decorative goods. Similarly, the customer 14911 purchases 60 CAKE CASES VINTAGE CHRISTMAS, SET OF 20 VINATEG CHRISTMAS NAPKINS, 36 DOLLIES VINTAGE CHIRSTMAS, SMALL POPCORN HOLDER, VINTAGE SNAP CARD. These products mostly fall into the category of Christmas-themed gift products and decorations, party supplies, children's products, lunchware, and decorations craft.

In addition, Customer 17450 frequently purchased products include CREAM HEART CARD HOLDER, HEART OF WICKER LARGE, WHITE HANGING HEART T-Light HOLDER, GREEN VINTAGE SPOT BREAKER and PINK VINATE SPOT BREAKER. Meanwhile, Customer 18102 shows a very similar purchasing pattern, with top products including CREAM HEART CARD HOLDER, BLACK HEART CARD HOLDER, WOOD BLACK BOARD AND WHITE FINISH, VINTAGE UNION JACK MEMOBOARD, DOORMAT HEARTS, and DOORMAT NEW ENGLAND. The similarity in their purchasing behavior highlights that these customers tend to buy items such as home decorations, kitchen and household accessories in moderate quantity. Therefore, this behavioral pattern reveals a trend in customer preferences towards the decoration products and crafts, which is a product category that the retailer should pay closer attention to.

These high-value customers are especially important because they generate a stable contribution to the overall revenue. Retaining such customers helps in ensuring financial stability to save the cost of attracting new customers. Therefore, the retailer should have strategies to keep these top five customers, as well as increase customer retention in general. This can be done by implementing initiatives such as customer loyalty programs, individually targeted promotions, special offers or vouchers. Conducting such loyalty

programs would generally be more cost effective than the actual process of attracting new customers. In addition, aligning marketing activities with these buying patterns would aid the retailer increase customer loyalty and generate high-margin revenue in the long-term. Finally, these insights from the customer analysis can further support the retailer to prioritize the inventory planning and ensure that the popular items are available.

Task 3: Global Market Analysis

The `Country_revenue_plot` demonstrates the total revenue across countries. The visualization shows that UK dominates in total revenue with 7,308,392 pounds compared to the other countries within the dataset. Following the UK are countries like Netherlands, EIRIE, Germany, France, and Australia, providing the next highest revenue. However, their level of revenue in comparison to the UK is still considerably lower, and most of the remaining countries generate relatively small amounts of revenue, resulting in a long-tail distribution with minimal revenue.

This revenue distribution highlights that the customer segments for the retailer are extremely concentrated in the domestic UK market, whilst international market contribution accounts for a small number of revenue contributions. The significant gap in the total revenue between domestic and international regions indicates the difference in demand between the two, challenges in market penetration, and limited accessibility of the retailer's products in the international markets. From the regional buying point of view, customers in the UK's market show stronger and more consistent purchasing power than the international market. This leads to the fact that the retailer currently relies heavily on the domestic market in the UK.

However, international markets such as Netherlands, EIRIE, Germany, France and Australia still indicate a positive pattern via their revenue contribution, representing potential opportunities for business expansion in those regions. Business penetration strategies like international marketing endeavors, enhancing shipping logistics, or region-specific offers can enhance customer satisfaction, customer retention, and sales in overseas markets.

Task 4: Temporal Analysis & Trends

The `Monthly_sales_plot` revealed the trend of monthly sales revenue in period of time from 2010-2011. Overall, the chart demonstrates the fluctuations of the revenue within a period of 12 months, which shows that instead of remaining stable, the sales trend witnesses a dramatic drop at the end of 2011.

At the beginning of the period, the revenue was rather stable as about 572,714 pounds during December 2010 to January 2011. However, the revenue declined dramatically to 447,137 pounds in the next month and reached at one of the lowest revenue levels in the time series chart. After this drop, the revenue recovered gradually in March 2011 and then witnessed another decrease in April of 2011. From May 2011, this trend repeated itself and started showing an upward movement in terms of total revenue. The most notable growth was between September 2011 and November 2011, when the revenue skyrocketed to 1,161,817 pounds. However, the revenue dropped significantly again in December 2011 to 518,193 pounds, suggesting seasonal fluctuations or incomplete data for the monthly period ahead.

From a business perspective, the rise in revenue experienced during the later part of the year may be due to seasonal buying behavior, as customer spending tends to rise toward the end of the year, possibly because of holiday shopping periods such as Halloween, Bonfire Night, Black Friday, Cyber Monday, Christmas, or even Boxing Day. However, the presence of low revenue months raises concerns about inventory costs and stock management issues, particularly when customers' buying power and customer demand decline. Therefore, the retailer may have to consider whether these items should keep on being stocked, or if promotional strategies are needed to stimulate the demand from customers. Gaining these temporal insights can support the retailer in making better decisions regarding inventory planning, market timing and anticipating any unexpected shifts in customer demand. This would ensure that popular products are adequately restocked during the peak demand periods while minimizing inventory and stockholding costs associated with consistently low-revenue products.